07/10/2026

Code-J_(Phase-4)

Aakash

Medical|IIT-JEE|Foundations

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MM : 720

Term Test for NEET-2028 (XI-Studying_P4)_TE-01J

Time : 180 Min.

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line
Chemistry: Some Basic Concepts of Chemistry, Structure of Atom
Botany: Cell: The Unit of Life, Cell Cycle & Cell Division
Zoology: Structural Organisation in Animals–Animal Tissues only, Biomolecules

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. The number of insignificant zeros in 0.006200 is
(1) Five (2) Three
(3) Two (4) Four

2. A physical quantity Z is related to four other quantities p, q, r and s as $Z = \frac{pq^2}{r^3\sqrt{s}}$. If the percentage errors in p, q, r and s are 1%, 2%, 3% and 2% respectively, then maximum percentage error in Z will be
(1) 14% (2) 10%
(3) 15% (4) 18%

3. Which of the following is most precise measurement of the diameter of a wire?
(1) 1.00 mm (2) 1.0 mm
(3) 1.00 cm (4) 1.0 cm

4. The radius of a sphere is (20.0 ± 0.2) cm. The percentage error in its surface area is
(1) 4% (2) 3%
(3) 2% (4) 1%

5. The number of significant figures in calculation of 7.3 + 3.028 are
(1) Two (2) Three
(3) Four (4) Five

6. A public park of square shape has measured edge length as (10.0 ± 0.1) m. The measured area of the park with error will be
(1) (100 ± 2) m^2 (2) (10 ± 2) m^2
(3) (100 ± 0.2) m^2 (4) (100 ± 0.1) m^2

7. The diameter of a cup is measured using a vernier caliper. The 20 vernier scale divisions are equal to 18 main scale divisions. If one division of main scale is x mm then least count of the instrument is
(1) $\left(\frac{x}{10}\right)$ mm (2) $\left(\frac{x}{20}\right)$ mm
(3) $\left(\frac{x}{100}\right)$ mm (4) $\left(\frac{x}{50}\right)$ mm

8. Which of the following measurement is least precise?
(1) 4.00 mm (2) 4.00 cm
(3) 4.00 m (4) 4.00 km

9. The given equation shows the variation of velocity (v) with time (t), $v = At^2 + \frac{Bt}{C+t}$, where A, B and C are constants. The dimensions of ABC will be
(1) $[M^0LT^{-2}]$ (2) $[M^0LT^{-3}]$
(3) $[M^0L^2T^{-3}]$ (4) $[M^0L^2T^{-2}]$

10. If unit of mass, length and time each is tripled, then the unit of force will
(1) Become 2 times (2) Remain same
(3) Become 8 times (4) Become 9 times

11. If a physical quantity Z depends on other two quantity P and Q according to given relation $Z = \frac{1}{(P+Q)}\left[P\tan\theta + Q\sin\theta\right]$, then select the **correct** option.
(1) The dimensions of P and Q can be different
(2) The dimensions of Z and P are same
(3) Z is a dimensionless physical quantity
(4) Both P and Q must be dimensionless

12. The time dependence of a physical quantity P is given by $P = P_0 e^{-\alpha t^3}$, where α is constant and t is time. The constant α
(1) Is dimensionless
(2) Has dimension of T^2
(3) Has dimensions T^3
(4) Has dimensions T^{-3}

13. The value of 1 kg/m^3 in CGS unit will be
(1) 10^{-6} g/cm^3
(2) 10^{-3} g/cm^3
(3) 10^{-2} g/cm^3
(4) 10^{-4} g/cm^3

14. If position x (in meter) depends upon time t (in second) as $x = at + bt^2$, then the SI unit of b will be
(1) m
(2) $\frac{m}{s}$
(3) $\frac{m}{s^2}$
(4) $\frac{m^2}{s}$

15. Taking into account of the significant figures, what is the value of 9.99 m – 0.0099 m?
(1) 9.98 m
(2) 9.980 m
(3) 9.9 m
(4) 9.9801 m

16. Dimensional formula of power is given by
(1) $[ML^2T^{-2}]$
(2) $[MLT^{-2}]$
(3) $[ML^2T^{-3}]$
(4) $[ML^{-2}T^{-3}]$

17. Which of the following has dimensions of $[ML^{-1}T^{-1}]$?
(1) Surface tension
(2) Coefficient of viscosity
(3) Bulk modulus
(4) Angular momentum

18. A vehicle travels half the distance with speed u and remaining distance with speed $3u$. Its average speed is
(1) $\frac{2u}{3}$
(2) $\frac{4u}{3}$
(3) $\frac{3u}{4}$
(4) $\frac{3u}{2}$

19. Position of an object is given as $x = 8t^2 + 5t$. The initial velocity of the object is
(1) Zero
(2) 13 m/s
(3) 8 m/s
(4) 5 m/s

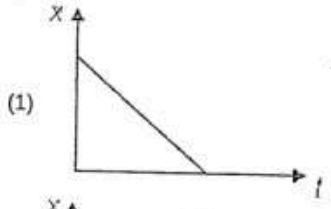
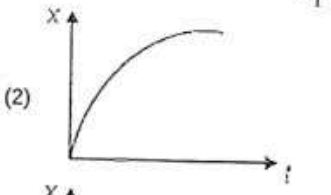
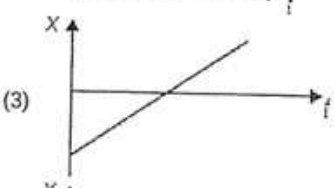
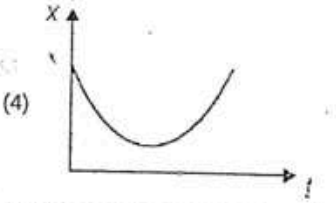
20. A particle moves along a circular path of radius 10 m. It completes half revolution in 5 s. The average velocity of particle will be
(1) 8 m/s
(2) 4 m/s
(3) $\frac{\pi}{2}$ m/s
(4) πm/s

21. A particle moves along a circular path of radius R. The ratio of distance and magnitude of displacement of particle in completing half revolution is
(1) $\pi : 2$
(2) $1 : 1$
(3) $\pi : \sqrt{2}$
(4) $\sqrt{2}\pi : 1$

22. If $y = x^2 - 2x + 3$ then what will be the minimum value of y?
(1) 1
(2) 3
(3) 2
(4) Zero

23. A car starts from rest and moves with constant acceleration a. The ratio of distances covered in the first second to the distance covered in third second is
(1) 1 : 2
(2) 2 : 1
(3) 3 : 1
(4) 1 : 5

24. The two ends of a train moving with constant acceleration pass a certain point with velocity 10 m/s and 30 m/s respectively. The velocity with which the middle point of the train passes the same point is
(1) 20 m/s
(2) 10 m/s
(3) 15 m/s
(4) $10\sqrt{5}$ m/s

25. The position-time (x-t) graph showing negative acceleration is
(1) 
(2) 
(3) 
(4) 

26. A car starts from rest and accelerates uniformly to a speed of 90 km/h in 10 s. The acceleration of particle is
(1) 1.5 m/s^2
(2) 3 m/s^2
(3) 2.5 m/s^2
(4) 9 m/s^2

27. A car moving along a straight highway with speed 144 km/h is brought to stop within a distance of 400 m with uniform retardation. How long does it take for the car to stop?
(1) 10 s
(2) 15 s
(3) 20 s
(4) 25 s

28. A ball projected vertically upward from the ground is at the same height at time t_1 and t_2. The total time taken by the ball to reach ground is [Neglect air resistance].
(1) $t_2 - t_1$
(2) $\sqrt{t_1 t_2}$
(3) $\frac{t_1 + t_2}{2}$
(4) $t_1 + t_2$

29. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion (A): If the body is slowing down then the angle between velocity and acceleration may be 180°.
Reason (R) : If the body is slowing down, the distance covered is negative.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true, (R) is false
(4) (A) is false, (R) is true

30. A train 100 m long moving at 30 m/s crosses another train 150 m long moving at 20 m/s in the opposite direction. The time taken by both of them to completely pass each other is
(1) 12.5 s
(2) 5 s
(3) 10 s
(4) 25 s

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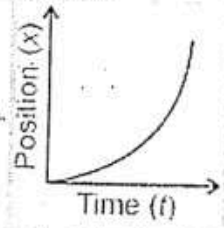
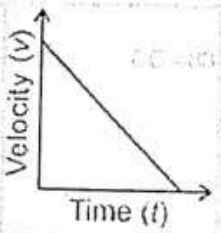
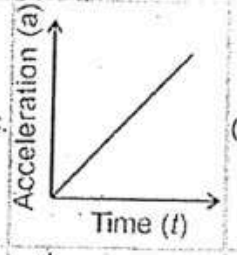
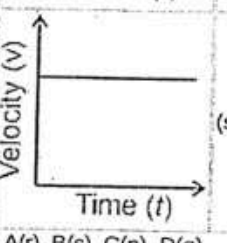
31. A metallic disc is being heated. Its area A (in m^2) at any time t (in second) is given by $A = 3t^2 + 5t + 9$. Rate of increases of area at $t = 3$ s is

(1) 9 m^2/s (2) 15 m^2/s
(3) 23 m^2/s (4) 25 m^2/s

32. Find the slope of the tangent to the curve $y = x^4 - 2x$ at $x = 2$

(1) 15 (2) 30
(3) 32 (4) 24

33. An object is moving along x-direction in four ways. Match the following in column-I with column-II.

Column-I	Column-II
A. 	(p) Non-uniform velocity
B. 	(q) Uniform velocity
C. 	(r) Positive (non-zero) acceleration
D. 	(s) Negative (non-zero) acceleration

(1) A(r), B(s), C(p), D(q)
(2) A(p, r), B(p, s), C(p, r), D(q)
(3) A(p, s), B(r), C(p, r), D(q)
(4) A(q, r), B(p, s), C(r), D(q)

34. Read the following statements carefully and mark the correct option.

Statement(A): Dimensionless quantities must have no unit.
Statement(B): Unitless quantities must have no dimension.
Statement(C): Dimensionless quantities may have unit.

(1) Statements (A) and (B) are correct
(2) Statements (B) and (C) are correct
(3) All statements (A), (B) and (C) are correct
(4) Only statement (A) is correct

35. Given are two statements:

Statement I: Area under velocity-time graph gives the distance travelled.
Statement II: For one-dimensional motion, a constant speed in an interval must have non-zero acceleration in that interval.

In the light of above statements, choose the most appropriate answer from the following

(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

36. A screw gauge has a least count of 5 μm and its circular scale is divided into 200 equal divisions. One main scale division is

(1) 0.005 m (2) 1 mm
(3) 1 cm (4) 0.0025 mm

37. Which of the following physical quantities has derived unit?

(1) Acceleration (2) Mass
(3) Current (4) Amount of substances

38. The damping force on an oscillator is directly proportional to the velocity. The units of the constant of proportionality is

(1) kg s^{-1} (2) kg s
(3) kg m s^{-1} (4) kg m s^{-2}

39. A student measures the distance traversed in free fall of a body, initially at rest in a given time. He uses this data to estimate g, the acceleration due to gravity. If the maximum percentage errors in measurement of the distance and the time are e_1 and e_2 respectively, the percentage error in the estimation of g is

(1) $e_2 - e_1$ (2) $e_1 + 2e_2$
(3) $e_1 + e_2$ (4) $e_1 - 2e_2$

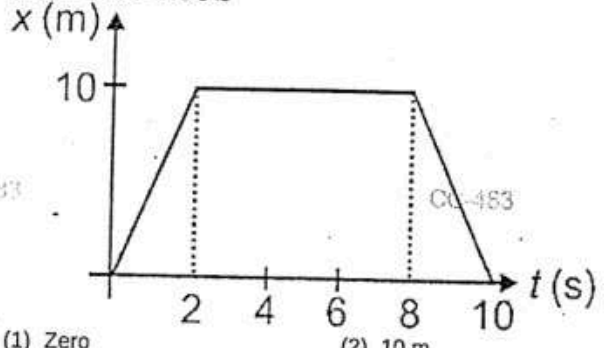
40. The pair of the quantities having same dimensions is

(1) Displacement, velocity (2) Time, frequency
(3) Wavelength, focal length (4) Force, acceleration

41. An object is moving with variable speed, then

(1) Its velocity may be zero
(2) Its velocity must be variable
(3) Its acceleration may be zero
(4) Its velocity must be constant

42. The position-time graph for a particle moving along a straight line is shown in figure. The total distance travelled by it in time $t = 0$ to $t = 10$ s is



(1) Zero (2) 10 m
(3) 20 m (4) 80 m

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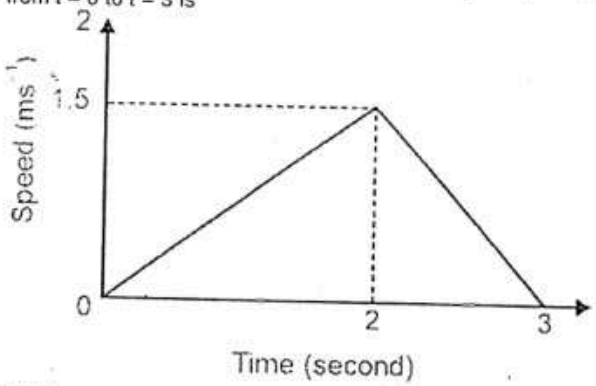
43. A body is projected vertically upward with speed 40 m/s. The distance travelled by body in the last second of upward journey is [take g = 9.8 m/s^2 and neglect effect of air resistance

(1) 4.9 m (2) 9.8 m
(3) 12.4 m (4) 19.6 m

44. A stone falls freely under gravity. It covers distances h_1, h_2 and h_3 in the first 5 seconds, the next 5 seconds and the next 5 seconds respectively. The relation between h_1, h_2 and h_3 is

(1) $h_1 = \frac{h_2}{3} = \frac{h_3}{5}$ (2) $h_2 = 3h_1$ and $h_3 = 3h_2$
(3) $h_1 = h_2 = h_3$ (4) $h_1 = 2h_2 = 3h_3$

45. The speed-time graph of a particle moving along a straight path is shown below. The distance traversed by the particle from $t = 0$ to $t = 3$ is



(1) $\frac{9}{2}$ m (2) $\frac{9}{4}$ m
(3) $\frac{10}{3}$ m (4) $\frac{10}{5}$ m

CHEMISTRY

46. The de-Broglie wavelength of an electron in 3rd Bohr orbit of He^+ ion is (a_0: radius of first Bohr orbit of H-atom)

(1) $2\pi a_0$ (2) $3\pi a_0$
(3) $6\pi a_0$ (4) πa_0

47. According to Bohr's theory angular momentum of an electron in 6th orbit of hydrogen atom is

(1) $2.5\frac{h}{\pi}$ (2) $6\frac{h}{\pi}$
(3) $3\frac{h}{\pi}$ (4) $\frac{2.5h}{2\pi}$

48. Match the List I containing multiples with their respective prefixes given in List II.

	List I		List II
(a)	10^{-15}	(i)	yocto
(b)	10^{-12}	(ii)	zepto
(c)	10^{-21}	(iii)	femto
(d)	10^{-24}	(iv)	pico

Choose the correct match:
(1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(4) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

49. The correctly reported answer of addition of 1.234 and 1.1 will have significant figures

(1) One (2) Two
(3) Three (4) Four

50. Suppose the elements A and B combine to form two compounds AB and AB_2. If 0.1 mole of AB weighs 6 g and 0.5 mole of AB_2 weighs 40 g, then atomic mass of A and B respectively are

(1) 20 u and 40 u (2) 40 u and 20 u
(3) 80 u and 40 u (4) 40 u and 80 u

51. If kinetic energy of a proton is increased nine times, the wavelength of the de-Broglie wave associated with it would become

(1) 3 times (2) 9 times
(3) $\frac{1}{3}$ times (4) $\frac{1}{9}$ times

52. For multielectronic species, the correct order of energy of the given orbitals is

(1) 3d > 4p > 5s > 5p (2) 5p > 5s > 4p > 3d
(3) 4p > 3d > 5p > 5s (4) 5s > 4p > 3d > 5p

53. Consider the following statements

(a) 1 amu = 1.66056×10^{-27} kg

(b) 1 mole of atomic H contains 6.022×10^{-23} hydrogen atoms

(c) SI unit of luminous intensity is candela

The correct statements are

(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (c) only (4) (a), (b) and (c)

54. Match List I with list II

List I (Metals)		List II (Work function)
(A) K	(i)	4.3 eV
(B) Cu	(ii)	2.42 eV
(C) Li	(iii)	4.8 eV
(D) Ag	(iv)	2.25 eV

Choose the correct answer from the options given below.
(1) (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)
(2) (A)-(iv), (B)-(iii), (C)-(ii), (D)-(i)
(3) (A)-(i), (B)-(iii), (C)-(ii), (D)-(iv)
(4) (A)-(iv), (B)-(iii), (C)-(i), (D)-(ii)

55. Consider the following statements about Dalton's atomic theory.

a. Matter consists of indivisible atoms.
b. Atoms of different elements differ in mass.
c. Compounds are formed when atoms of different elements combine in a fixed ratio.
d. Atoms are neither created nor destroyed in a chemical reaction.

Correct statements are

(1) a and b only (2) b and c only
(3) c and d only (4) a, b, c and d

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56. If percentage of Zn (atomic mass = 65.38 u) in a biomolecule is 0.75% by weight, then the minimum molar mass (in g mol^{-1}) of biomolecule will be
(1) 10243.45 (2) 9483.29
(3) 8012.12 (4) 8717.33

57. The concentration term that gets affected by temperature is
(1) Molality (2) Normality
(3) Mole fraction (4) (wt/wt) percent

58. Which among the following is incorrect observation during the photoelectric effect?
(1) No time lag between the striking of light and ejection of electrons
(2) The number of electrons ejected does not depend upon the frequency of light used
(3) The kinetic energy of the photoelectrons depends upon the frequency of light used
(4) If the frequency of light used is lesser than the threshold frequency only then electrons are ejected from metal

59. If the molality of the aqueous solution is 0.111, then the mole fraction of solute will be
(1) 0.02 (2) 0.002
(3) 0.2 (4) 0.1

60. From 220 mg of CO_2, 10^{21} molecules of CO_2 are removed. Number of remaining molecules of CO_2 will be
(1) 3.011×10^{21} (2) 2.011×10^{21}
(3) 1.011×10^{21} (4) 6.022×10^{21}

61. Consider the following statements:
Statement I: Precision refers to the agreement of a particular value to the true value of the result.
Statement II: Accuracy refers to closeness of various measurements for the same quantity.
In light of above statements, choose the correct answer.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are incorrect
(4) Both statement I and statement II are correct

62. Mass of 80% pure $CaCO_3$ required to neutralise 100 mL of 0.5 M HCl solution will be
(1) 2.06 g (2) 3.12 g
(3) 4.15 g (4) 4.50 g

63. The mass of precipitate (Ag_2CO_3) formed, when 50 ml of 21.2% (w/v) solution of Na_2CO_3 is mixed with 100 ml of 17% (w/v) $AgNO_3$ solution is (molar mass: $AgNO_3$ = 170 g mol^{-1}, Na_2CO_3 = 106 g mol^{-1} and Ag_2CO_3 = 276 g mol^{-1})
(1) 25.9 g (2) 16.7 g
(3) 31.4 g (4) 13.8 g

64. Values of all four quantum number for the last electron of potassium atom are (Atomic number of K = 19)
(1) n = 4, l = 0, m = 0, s = $\frac{+1}{2}$
(2) n = 4, l = 1, m = 0, s = $\frac{-1}{2}$
(3) n = 3, l = 1, m = 0, s = $\frac{+1}{2}$
(4) n = 3, l = 0, m = 0, s = $\frac{-1}{2}$

65. The lightest particle among the following is
(1) Electron (2) Proton
(3) Neutron (4) α-particle

66. A 100 watt bulb emits monochromatic light of wavelength 400 nm. The number of photons emitted per second by the bulb is
(1) 1×10^{15} (2) 2×10^{20}
(3) 2×10^{10} (4) 8×10^{20}

67. The number of lines obtained in UV region of hydrogen spectrum when electron jumps from 6th excited state to 1st excited state is/are:
(1) 5 (2) Zero
(3) 4 (4) 3

68. Consider the combustion reaction of 8 g of methane with 22.4 L oxygen gas at STP.
Statement I: Oxygen will get completely consumed.
Statement II: If the mass of both reactants is reduced to half, then the mass of CO_2 obtained will also get reduced to half.
In the light of above statements, choose the **correct** option.
(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Statement I is correct but Statement II is incorrect

69. If the density of methanol is 0.793 Kg L^{-1}. The volume of methanol needed for making 3 L of 0.3 M aqueous solution is
(1) 36.32 mL (2) 28.34 mL
(3) 32.36 mL (4) 38.58 mL

70. Given below are two statements
Statement I: The shapes of d_{xy}, d_{yz}, d_{xz} and d_{z^2} orbitals are similar to each other
Statement II: All five 3d orbitals are equivalent in energy in free state.
In the light of above statements, choose the correct answer from the options given below
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

71. The energy of 2*s* and 2*p* orbitals is the same for
(1) Li (2) Li^{2+}
(3) Be^{2+} (4) Be

72. Rutherford's experiment, which established the nuclear model of the atom, used a beam of
(1) Helium atoms, which impinged on a metal foil and got scattered
(2) β-particles, which impinged on a metal foil and got absorbed
(3) γ-rays which impinged on a metal foil and ejected electrons
(4) Helium nuclei, which impinged on a metal foil and got scattered

73. Total number of electrons present in *p*-orbitals of a sulphur atom is
(1) 10 (2) 12
(3) 6 (4) 8

74. If an element, X has the following isotopic composition: ^{54}X : 70%; ^{53}X : 20%; ^{55}X : 10%. The weighted average atomic mass of the naturally occurring element X will be
(1) 53.1 u (2) 54.1 u
(3) 53.9 u (4) 54.9 u

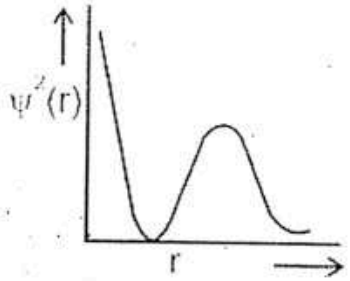
75. Total number of exchanges possible for $3d^4$ configuration is
(1) Four (2) Two
(3) Six (4) Eight

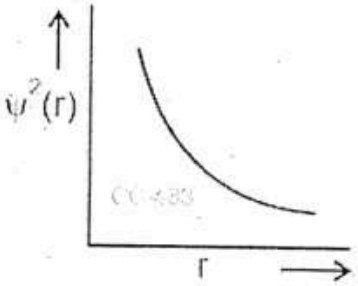
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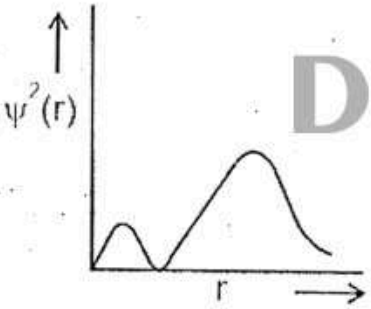
76. The ratio of specific charge of an electron to that of a proton is

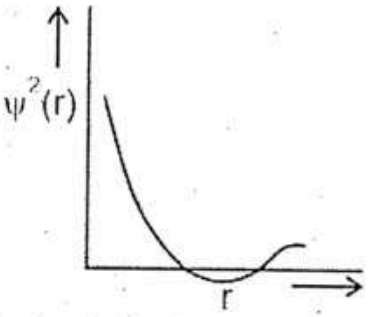
(1) 1 : 1 (2) 1837 : 1
(3) 1 : 1837 (4) 2 : 1

77. The probability density curve for 2s electron appears like

(1) 

(2) 

(3) 

(4) 

78. Match column I with column II

Column I (Orbitals)	Column II (no. of angular nodes)
a. 1s	(i) 1
b. 2p	(ii) 3
c. 4d	(iii) 0
d. 4f	(iv) 2

The correct option is

(1) a(iv), b(iii), c(ii), d(i) (2) a(iii), b(i), c(ii), d(iv)
(3) a(i), b(ii), c(iii), d(iv) (4) a(iii), b(i), c(iv), d(ii)

79. A 3.3 mg ball is moving with a speed of 200 m/s. The de Broglie wavelength associated with it will be ($h = 6.6 \times 10^{-34}$ Js)

(1) 10^{-27} m (2) 10^{-30} m
(3) 10^{-24} m (4) 10^{-33} m

80. Consider the following statements:

(i). Number of atoms in 80 u of calcium is 2

(ii). Number of atoms in 2240 mL of CO_2 at STP is 6.022×10^{22}

(iii). A sample of drinking water contains 10 ppm (by mass) of impurities. The percentage by mass of impurities in the sample is 10^{-3}

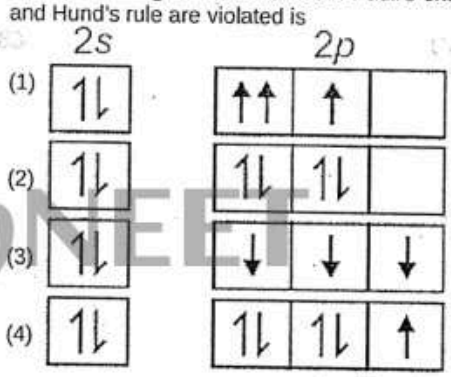
Choose the correct statement(s)

(1) (i) & (ii) only (2) (i) only
(3) (ii) & (iii) only (4) (i) & (iii) only

81. If the uncertainty in the position of electron is zero, the uncertainty in its momentum would be

(1) Zero (2) Greater than $\frac{h}{4\pi}$
(3) Less than $\frac{h}{4\pi}$ (4) Infinite

82. The orbital diagram in which both Pauli's exclusion principle and Hund's rule are violated is

(1) 

(2)

(3)

(4)

83. The frequency of a wave is 6×10^{15} s^{-1}. Its wave number would be

(1) 10^5 cm^{-1} (2) 2×10^7 m^{-1}
(3) 2×10^7 cm^{-1} (4) 2×10^5 m^{-1}

84. In a hydrogen atom, if the energy of electron in the ground state is –x eV., then that in the 2nd excited state of He^+ is

(1) –x eV (2) $-\frac{4}{9}$x eV
(3) +2x eV (4) $-\frac{9}{4}$x eV

85. A certain metal when irradiated with light ($\nu = 3.2 \times 10^{16}$ Hz) emits photo electrons with twice kinetic energy as did photo electrons when the same metal is irradiated by light ($\nu = 2.0 \times 10^{16}$ Hz). Calculate ν_0 of electron?

(1) 1.2×10^{14} Hz (2) 8×10^{15} Hz
(3) 1.2×10^{16} Hz (4) 4×10^{12} Hz

86. The amount of zinc required to produce 1.12 mL of H_2 at STP on treatment with dilute HCl will be (Molar mass of Zn = 65.38 g/mol)

(1) 65 g (2) 0.065 g
(3) 32.5×10^{-4} g (4) 6.5 g

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87. The percentage of hydrogen in water and hydrogen peroxide is 11.1 and 5.9 respectively. These figures illustrate
(1) Law of multiple proportions
(2) Law of conservation of mass
(3) Law of constant proportions
(4) Law of combining volumes

88. Which has the maximum number of molecules among the following?
(1) 8 g H_2 (2) 64 g SO_2
(3) 44 g CO_2 (4) 48 g O_3

89. Equivalent mass of a metal is 12. Hence, equivalent mass of its oxide is
(1) 24 (2) 28
(3) 20 (4) 34

90. What is the ratio of empirical formula mass to molecular formula mass of benzene?
(1) 1 : 6 (2) 2 : 3
(3) 6 : 1 (4) 3 : 2

BOTANY

91. Match the following columns and select the **correct** option.

Column-I	Column-II
a. Antonie Von Leeuwenhoek	(i) Discovered the cell from a thin slice of cork
b. Robert Brown	(ii) First saw and described a live cell
c. Robert Hooke	(iii) Gave cell theory its final shape
d. Rudolf Virchow	(iv) Discovered nucleus

(1) a(ii), b(iv), c(i), d(iii) (2) a(i), b(ii), c(iii), d(iv)
(3) a(ii), b(i), c(iii), d(iv) (4) a(iii), b(ii), c(i), d(iv)

92. Read the given statements and select **correct** option.
A: The *cis* and the *trans* face of Golgi apparatus are entirely different and are not interconnected.
B: Golgi apparatus remains in close association with ER.
C: Cisternae in Golgi apparatus are stacked parallel to each other.
(1) Only statement A is incorrect.
(2) Statements A and C are correct.
(3) All statements A, B and C are correct.
(4) Statements A and B are incorrect.

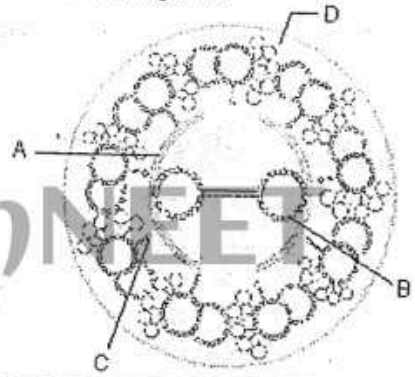
93. Read the following statements and select the **correct** option.
Statement I: None of the organelles is common among prokaryotic and eukaryotic cell.
Statement II: Many bacteria have small circular DNA outside the genomic DNA which confers certain unique phenotypic character to such bacteria
(1) Only statement I is correct
(2) Only statement II is correct
(3) Both statements I and II are correct
(4) Both statements I and II are incorrect

94. Algal cell wall is composed of
(a) Galactans
(b) Calcium carbonate
(c) Mannans
(d) Cellulose
The **correct** one(s) is/are
(1) All (a), (b), (c) and (d) (2) Only (d)
(3) Only (a), (c) and (d) (4) Only (a) and (c)

95. The microtubules are
(1) Found in both prokaryotic and eukaryotic cells
(2) Found only in prokaryotic cells and made up of 13 parallel protofilaments
(3) Hollow, unbranched cylindrical structure found only in eukaryotic cells
(4) Solid, unbranched, rod-like fibrils of indefinite length found only in eukaryotic cells

96. Select the **correct** statement about eukaryotic cilia and flagella.
(1) Cilia and flagella give a cartwheel appearance
(2) Axoneme is made up of nine microtubule triplets at periphery
(3) They show '9 + 0' microtubular organisation in axoneme
(4) Both the cilium and flagellum emerge from centriole-like structure called the basal bodies

97. Identify the structures marked as A, B, C and D in the given section of a flagellum.



	A	B	C	D
(1)	Central bridge	Radial spoke	Inter-doublet bridge	Plasma membrane
(2)	Plasma membrane	Central microtubule	Radial spoke	Central sheath
(3)	Central sheath	Central bridge	Radial spoke	Plasma membrane
(4)	Central sheath	Central microtubule	Radial spoke	Plasma membrane

(1) (1) (2) (2)
(3) (3) (4) (4)

98. Cell wall is
(1) A living rigid structure found in the plant cells
(2) The outer covering of the plasma membrane of all eukaryotic organisms
(3) Capable of growth and expansion in young plant cell
(4) A barrier to macromolecule but does not protect from infection.

99. A unique feature of prokaryotic cell is the presence of
(1) 70S ribosomes (2) Chitinous cell wall
(3) Naked genetic material (4) Double stranded DNA

100. Nucleolus is the site of the synthesis of
(1) rRNA (2) tRNA
(3) DNA (4) mRNA

DreamNEET

101. Which layer of the cell envelope prevents the bacterium from bursting or collapsing?
(1) Plasma membrane (2) Glycocalyx
(3) Cell wall (4) Capsule

102. Choose the **incorrect** statement about the main arena of cellular activities of a cell.
(1) It is a semi-fluid matrix
(2) It is found in eukaryotes only
(3) It occupies the volume of cell
(4) It contains enzymes and cell organelles

103. Choose the **incorrect** statement about mitochondria.
(1) Site of anaerobic respiration
(2) Produce cellular energy in the form of ATP
(3) Are called 'power house' of cell
(4) Possess single circular DNA molecule in their matrix

104. Which of the following is not the function of mesosome?
(1) Cellular respiration (2) DNA replication
(3) Help in cell division (4) Photosynthesis

105. Kinetochores are directly associated with
(1) Primary constriction (2) Centrosome
(3) Secondary constriction (4) Satellite

106. Fluid nature of cell membrane is related to some functions(s) like
(a) Cell growth
(b) Protection of cells from mechanical injury
(c) Formation of intercellular junctions
(d) Endocytosis
(1) Only (a) (2) Only (a) and (b)
(3) Only (a), (c) and (d) (4) All (a), (b), (c) and (d)

107. Which of the following is associated with detoxification of drugs and muscle contraction by the release and uptake of Ca^{2+} ions?
(1) Golgi complex (2) RER
(3) SER (4) Free ribosomes

108. Movement of ions through Na^+/K^+ pump in animal cells
(1) Is a kind of passive transport
(2) Is against their concentration gradient
(3) Does not require energy
(4) Occurs via simple diffusion

109. Special membranous extension present in the form of vesicles, tubules and lamellae in Gram positive bacteria is called
(1) Glycocalyx (2) Mesosome
(3) Capsule (4) Slime layer

110. Read the following **Assertion (A)** and **Reason (R)** statements and choose the correct option
Assertion (A) : Unicellular organisms are capable of independent existence.
Reason (R) : Unicellular organisms perform the essential functions of life on their own.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

111. State **true (T)** or **false (F)** for the given statements and select the **correct** option.
A) After meiosis I, the resultant daughter cells have twice the number of chromatids as well as the amount of DNA in comparison to haploid gametes.
B) Reorganisation of all cell components take place during the G_1 phase.
C) The synapsed homologous chromosomes, become clearly visible as tetrad at pachytene stage.
D) Anaphase and anaphase II are similar in a single individual in terms of the genetic variability of chromatids.

	A	B	C	D
(1)	T	F	T	F
(2)	F	T	F	T
(3)	T	T	F	F
(4)	T	F	F	T

(1) (1) (2) (2)
(3) (3) (4) (4)

112. Which phase of the cell cycle lasts more than 95% of the duration of cell cycle?
(1) M-phase (2) G_0-phase
(3) Interphase (4) Synthetic phase

113. Nucleolus, Golgi complex and ER reappear during
(1) Prophase (2) Metaphase
(3) Telophase (4) Anaphase

114. Arrange the following events of mitosis in the **correct** sequence.
(a) Condensation of genetic material
(b) Splitting of centromere
(c) Reassembly of nuclear envelope
(d) Arrangement of chromosomes at equator
(e) Segregation of sister chromatids
(1) (a) → (d) → (b) → (e) → (c)
(2) (a) → (b) → (d) → (e) → (c)
(3) (d) → (b) → (a) → (c) → (e)
(4) (d) → (b) → (c) → (a) → (e)

115. Choose **incorrect** option w.r.t. cell cycle
(1) Cell growth in terms of cytoplasmic increase occurs continuously
(2) DNA synthesis occurs during specific stage
(3) Events are not under genetic control
(4) Cell duplicates its genome once

116. Best stage to study the shape of chromosomes is
(1) Anaphase (2) Metaphase
(3) Prophase (4) Telophase

117. The first stage of karyokinesis of mitosis is
(1) Anaphase (2) Telophase
(3) Metaphase (4) Prophase

118. The beginning of diplotene stage is recognised by
(1) Terminalisation of chiasmata completely
(2) Dissolution of synaptonemal complex
(3) Disappearance of nuclear membrane completely
(4) Synapsis of chromosomes

119. Read the following assertion (A) and reason (R) statements and select the correct option.
Assertion (A): The chromosomes are spread through the cytoplasm of the cell during metaphase.
Reason (R): The complete disintegration of the nuclear envelope marks the start of the metaphase.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

120. Stage between the meiosis-I and meiosis-II is
(1) Characterised by DNA as well as centriole duplication
(2) Called cytokinesis
(3) Characterised by polymerisation of deoxyribonucleotides
(4) Short lived than preparatory phase of the cell cycle

121. Presence of which of the following structures is common in animal and higher plant cells?
(1) Mitochondria (2) Plastids
(3) Centrioles (4) Large central vacuole

122. Which of the following is not the part of endomembrane system?
(1) Lysosome (2) Golgi body
(3) Mitochondria (4) ER

123. Read the given statements and select the **correct** option.
Statement A : The plants can show mitotic division in both haploid and diploid cells.
Statement B : The G_1 phase of interphase is immediately followed by the first stage of karyokinesis of mitosis, i.e., Prophase.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements are correct
(4) Both statements are incorrect

124. Anaphase II differs from Anaphase of mitosis in
(1) Showing separation of chromatids
(2) Its occurrence in haploid cells only
(3) Involvement of splitting of centromere
(4) Showing the movement of chromatids towards the poles

125. The role of meiosis is to
(1) Form genetically identical gametes
(2) Reduce the number of chromosomes by half in the gametes
(3) Restore the original diploid number of chromosomes in the zygote
(4) Bring two non-homologous chromosomes together for recombination

126. An elaborate network of filamentous proteinaceous structure present in the cytoplasm which helps in the maintenance of cell shape is called
(1) Thylakoid (2) Endoplasmic reticulum
(3) Plasmalemma (4) Cytoskeleton

127. Which structure is/are considered as semi-autonomous organelle?
(1) Ribosome
(2) Golgi body
(3) Mitochondria and chloroplast
(4) Mitochondria only

128. Plasma membrane is
(1) Semipermeable and symmetric
(2) Selectively permeable, elastic and asymmetric
(3) Permeable and asymmetric
(4) Selective permeable with monolayer phospholipids

129. The larger sub-unit of a ribosome is found to contain 28S, 5.8S and 5S types of RNA. This ribosome is found in
(1) Bacterium (2) Mitochondrion
(3) Animal cell (4) Chloroplast

130. The smallest cell is
(1) Ostrich egg (2) Mycoplasma
(3) WBC (4) Liver cell

131. A single cell containing large number of nuclei is called
(1) Syncytium (2) Cell plate
(3) Monad (4) Bivalent

132. Spireme stage of chromosomes is associated with
(1) Prophase (2) Anaphase
(3) Metaphase (4) Telophase

133. Bivalent chromosomes align themselves at the equator during
(1) Metaphase I (2) Prophase I
(3) Metaphase II (4) Anaphase II

134. When synapsis is complete all along the chromosome, the cells are said to have entered a stage of prophase I, where exchange of genetic material takes place between homologous chromosomes. The stage is called
(1) Zygotene (2) Pachytene
(3) Diplotene (4) Diakinesis

135. Temporarily suspended stage of diplotene during meiosis-I is called
(1) Leptotene (2) Diakinesis
(3) Dictyotene (4) Pachytene

ZOOLOGY

136. Which of the following is an acidic amino acid?
(1) Phenylalanine (2) Tyrosine
(3) Aspartate (4) Glycine

137. How many number of amino acids and peptide bonds are present in a tripeptide respectively?
(1) 3, 3 (2) 3, 2
(3) 2, 3 (4) 2, 2

138. Choose the odd one w.r.t. secondary metabolites which impart colour as well.
(1) Carotenoid (2) Curcumin
(3) Morphine (4) Anthocyanin

139. Enzymes which catalyse oxidation-reduction reactions involving transfer of electrons/H^+ from one molecule to another are called
(1) Transferases (2) Hydrolases
(3) Dehydrogenases (4) Lyases

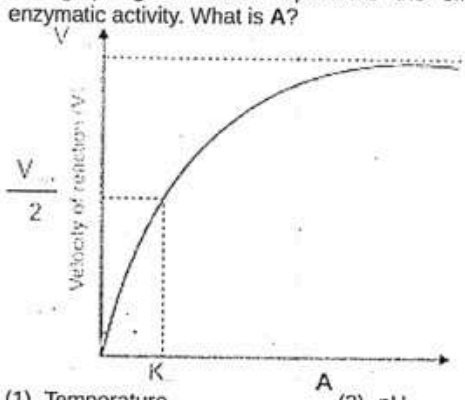
140. Bone is mainly composed of
(1) Calcium carbonate (2) Calcium phosphate
(3) Calcium sulphate (4) Calcium chloride

141. Fusiform shaped, unstriped and uninucleate condition are **not** the characteristic features of muscle fibres present in the wall of
(1) Stomach (2) Small intestine
(3) Arteries (4) Human heart

Dream NEET

142. Which of the following is **not** a characteristic of glycogen?
(1) Heteropolysaccharide
(2) Homopolysaccharide
(3) Branched chain molecule
(4) Stored in liver and muscle

143. The graph given below represents the effect of **A** on enzymatic activity. What is **A**?



(1) Temperature
(2) pH
(3) Substrate concentration
(4) Product concentration

144. Guanosine is a
(1) Nucleoside
(2) Nitrogenous base
(3) Nucleotide
(4) Pyrimidine

145. Structure that is **not** related to cyton of a neuron is
(1) Nissl's granules
(2) Nucleus
(3) Dendrites
(4) Node of Ranvier

146. Which of the following is a non-reducing sugar?
(1) Sucrose
(2) Maltose
(3) Lactose
(4) Glucose

147. Synthesis of _____ does not require the formation of glycosidic bond.
Choose the option that fills the blank correctly.
(1) Ribose
(2) Adenylic acid
(3) Sucrose
(4) Cellulose

148. A double stranded DNA molecule contains 20% guanine. What will be the expected percentage of other three nitrogenous bases, *i.e.*, cytosine, adenine and thymine represented as C, A and T?
(1) C - 20%, A - 30%, T - 30%
(2) C - 30%, A - 20%, T - 30%
(3) A - 40%, T - 20%, C - 20%
(4) A - 20%, T - 20%, C - 40%

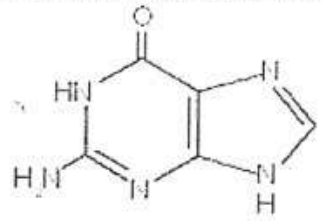
149. All involuntary muscles
(1) Get fatigued easily
(2) Are striated
(3) Are uninucleated
(4) Have unbranched fibres

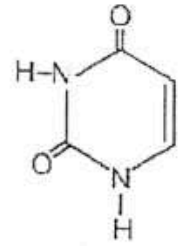
150. In primary structure of protein, the N-terminal amino acid refers to ___ amino acid. Choose the option that fills the blank **correctly.**
(1) First
(2) Second
(3) Last
(4) Second to last

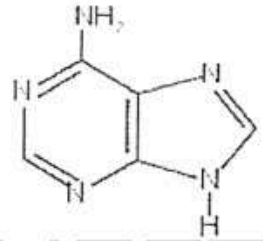
151. Most abundant structural polysaccharide in nature is a secondary metabolite named as
(1) Glycogen
(2) Starch
(3) Insulin
(4) Cellulose

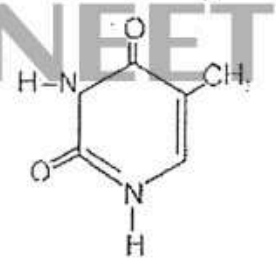
152. Choose the **correct** option representing set of connective tissues only.
(1) Blood, lymph, muscle
(2) Bone, blood, muscle
(3) Tendons, bone, blood
(4) Lymph, blood, astrocytes

153. Among the following nitrogenous bases, select a purine that can form two hydrogen bonds with a pyrimidine.

(1) 

(2) 

(3) 

(4) 

154. Bone is surrounded by a tough sheath of connective tissue called
(1) Perichondrium
(2) Periosteum
(3) Sarcolemma
(4) Axolemma

155. Enzymes get permanently damaged and loose catalytic power at
(1) Low temperature
(2) Optimum temperature
(3) High temperature
(4) Optimum pH

156. Brush border simple cuboidal epithelium is present in the inner surface of
(1) Trachea
(2) Bronchioles
(3) PCT of nephron
(4) Small intestine

157. Primary structure of proteins is due to the presence of
(1) Hydrogen bonds
(2) Peptide bonds
(3) Disulphide
(4) Ionic bonds

158. The type of epithelium that is composed of a single layer of cells perform all of the given functions, **except**
(1) Lining for body cavities
(2) Forming diffusion boundary
(3) Secretion
(4) Protection from chemical stresses

DreamNEET

159. Biomolecules present in living organisms are categorised into filtrate and retentate fractions during their biochemical analysis. Among these two categories which one contains cellulose?
(1) Water soluble fraction (2) Both filtrate and retentate
(3) Filtrate only (4) Retentate fraction

160. Heparin and histamine are released by
(1) Fibroblasts (2) Mast cells
(3) Osteocytes (4) Adipocytes

161. Inulin is obtained from tuberous root of Dahlia and is considered as a
(1) Homopolymer
(2) Heteropolymer
(3) Sulphur containing polymer
(4) Nitrogen containing polymer

162. Muscles present in walls of blood vessels are
(1) Striated and voluntary
(2) Non-striated and involuntary
(3) Striated and involuntary
(4) Non-striated and voluntary

163. Consider the given secretions :-
(a) Earwax
(b) Milk
(c) Enzymes
(d) Mucus
Which of the above secretions are not released from endocrine glands?
(1) a and b only (2) b and c only
(3) c and d only (4) a, b, c and d

164. **Incorrect** match among the following is
(1) Haem –Prosthetic group of catalase
(2) Zn^{2+} – Co-factor for carboxypeptidase
(3) NADP – Co-factor that contains vitamin B_9
(4) FMN – Binds to apoenzyme transiently

165. The fibrous tissue which connects the two bone is
(1) Connective tissue (2) Tendon
(3) Ligament (4) Adipose tissue

166. Trihydroxypropane is predominantly a component of
(1) Lipids (2) Nucleic acids
(3) Proteins (4) Carbohydrates

167. The type of cell junction found in epithelium which helps primarily to stop substances from leaking across the tissue is
(1) Tight junction (2) Gap junction
(3) Adhering junction (4) Desmosome

168. Neuroglia make up more than _____ the volume of neural tissue in our body.
Select the option which fills the blank **correctly**.
(1) One-fifth (2) One-fourth
(3) One-third (4) One-half

169. Concanavalin A is a/an
(1) Alkaloid (2) Terpenoid
(3) Lectin (4) Pigment

170. Ciliated cells are found in
(1) Bronchus (2) Pancreas
(3) Liver (4) Alveoli

171. Cells of squamous epithelium are
(1) Tall and slender with basal nuclei
(2) Tall with cilia on their free surface
(3) Flattened with central nuclei
(4) Cube like with central nuclei

172. Intercalated discs are specialised regions of cell membrane of two adjacent muscle fibres and are present in
(1) Smooth muscle (2) Cardiac muscle
(3) Skeletal muscle (4) Voluntary muscle

173. The packing cells around the neurons in neural tissue are called
(1) Mast cells (2) Histiocytes
(3) Neuroglial cells (4) Fibroblasts

174. Muscles involved in the movement of arm possess muscle fibres that are
(1) Striated (2) Unstriated
(3) Uninucleated (4) Fusiform in shape

175. Simple epithelium is made of
(1) Non-cellular layer of hyaluronic acid
(2) Many layer of cells
(3) Loosely arranged cells
(4) Compactly packed single layer of cells

176. Antibodies that help to fight infectious agents are classified under the category of
(1) Lipids (2) Homopolysaccharides
(3) Proteins (4) Nucleic acids

177. Beta-pleated structure in a protein is associated with which level of structural organisation?
(1) Primary (2) Secondary
(3) Tertiary (4) Quaternary

178. Which of the following statements is **incorrect**?
(1) Lipids are strictly macromolecules
(2) Palmitic acid has 16 carbons including carboxyl carbon
(3) Oils have low melting point and hence remain as liquid in winters
(4) Arachidonic acid is an unsaturated fatty acid

179. Select the **correct** match between column I and column II.

	Column I		Column II
a.	Cellulose	(i)	Storage homo-polysaccharide
b.	Chitin	(ii)	Structural homo-polysaccharide
c.	Inulin	(iii)	Branched polymer of glucose
d.	Glycogen	(iv)	Exoskeleton of arthropods
		(v)	Cannot trap I_2 molecules

Choose the **correct** option.
(1) a(v), b(iv), c(i), d(iii) (2) a(ii), b(iii), c(i), d(iv)
(3) a(v), b(iv), c(ii), d(i) (4) a(ii), b(iv), c(iii), d(i)

180. Select the **incorrect** match.
(1) Lecithin – Phosphoglyceride
(2) Competitive inhibition – Decrease in K_m and V_{max} value
(3) Heterocyclic ring – Adenine and guanine
(4) Coenzymes – Vitamin derivatives

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07/10/2026

CODE-J
Phase-4

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TERM EXAM FOR NEET-2028 (XI STUDYING)

MM: 720 **TE-01** **Time: 180 Mins.**

Answers

1. (2)	37. (1)	73. (1)	109. (2)	145. (4)
2. (3)	38. (1)	74. (3)	110. (1)	146. (1)
3. (1)	39. (2)	75. (3)	111. (1)	147. (1)
4. (3)	40. (3)	76. (2)	112. (3)	148. (1)
5. (2)	41. (2)	77. (1)	113. (3)	149. (3)
6. (1)	42. (3)	78. (4)	114. (1)	150. (1)
7. (1)	43. (1)	79. (2)	115. (3)	151. (4)
8. (4)	44. (1)	80. (4)	116. (1)	152. (3)
9. (3)	45. (2)	81. (4)	117. (4)	153. (3)
10. (2)	46. (2)	82. (1)	118. (2)	154. (2)
11. (3)	47. (3)	83. (2)	119. (1)	155. (3)
12. (4)	48. (4)	84. (2)	120. (4)	156. (3)
13. (2)	49. (2)	85. (2)	121. (1)	157. (2)
14. (3)	50. (2)	86. (3)	122. (3)	158. (4)
15. (1)	51. (3)	87. (1)	123. (1)	159. (4)
16. (3)	52. (2)	88. (1)	124. (2)	160. (2)
17. (2)	53. (3)	89. (3)	125. (2)	161. (1)
18. (4)	54. (2)	90. (1)	126. (4)	162. (2)
19. (4)	55. (4)	91. (1)	127. (3)	163. (4)
20. (2)	56. (4)	92. (1)	128. (2)	164. (3)
21. (1)	57. (2)	93. (2)	129. (3)	165. (3)
22. (3)	58. (4)	94. (1)	130. (2)	166. (1)
23. (4)	59. (2)	95. (3)	131. (1)	167. (1)
24. (4)	60. (2)	96. (4)	132. (1)	168. (4)
25. (2)	61. (3)	97. (4)	133. (1)	169. (3)
26. (3)	62. (2)	98. (3)	134. (2)	170. (1)
27. (3)	63. (4)	99. (3)	135. (3)	171. (3)
28. (4)	64. (1)	100. (1)	136. (3)	172. (2)
29. (3)	65. (1)	101. (3)	137. (2)	173. (3)
30. (2)	66. (2)	102. (2)	138. (3)	174. (1)
31. (3)	67. (2)	103. (1)	139. (3)	175. (4)
32. (2)	68. (1)	104. (4)	140. (2)	176. (3)
33. (2)	69. (1)	105. (1)	141. (4)	177. (2)
34. (2)	70. (4)	106. (3)	142. (1)	178. (1)
35. (2)	71. (2)	107. (3)	143. (3)	179. (1)
36. (2)	72. (4)	108. (2)	144. (1)	180. (2)

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